

Detecting GDPR Policy Deviations with an LLM-Based Pipeline

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Abstract

Manually checking a multi-page company data protection policy against the hundreds of obligations of the General Data Protection Regulation (GDPR) is error-prone and does not scale. Sai et al. [2023] introduced the RCASR taxonomy and a semi-automatic NLP pipeline for this task, but their approach depends on signal-word filters, manually tuned similarity thresholds, and a syntactic step that splits each constraint into responsibility, task and data slots via part-of-speech tagging and dependency parsing, components that are brittle on free-form policy text. We replace these with three Large Language Model (LLM) steps: a contextual obligation extractor with a ± 5 -sentence window, an LLM-as-judge that confirms Legal-BERT retrieval candidates, and a zero-shot deviation classifier covering six RCASR types. The pipeline is model-agnostic: the reported results use OpenAI GPT-4o-mini, and the same prompts run unchanged against Qwen-3.5-9B served locally via Ollama, allowing a fully on-premise deployment that requires no external API. Evaluated against gold standards of seventeen controlled deviations injected into the real privacy policies of Hetzner, Zalando, and Trade Republic, the pipeline reaches $F_1 = 34.7\%$ at recall 49.0%, with the *execution_style* type best detected ($F_1 = 57.1\%$). The result demonstrates that an off-the-shelf instruction-tuned LLM is competitive with the original semi-automatic system on recall while eliminating every manually-supervised step: the pipeline runs end-to-end without operator intervention.

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1. Introduction

Recent regulatory enforcement leaves no doubt that GDPR compliance is a first-order business risk. The Irish Data Protection Commission’s €1.2 billion fine against Meta in May 2023 for unlawful transfers of European personal data to the United States [European Data Protection Board, 2023] stands as the largest single GDPR penalty to date and illustrates the scale of exposure for any company that processes European user data. Compliance with the GDPR is not a one-off audit task, either: the regulation contains hundreds of operational obligations distributed across dozens of articles, and a multi-page company data protection policy must be checked against every applicable obligation. Doing so by hand does not scale to the thousands of policies maintained by EU-active companies.

Sai et al. [2023] introduced an explicit framework for this matching problem, the Regulatory Constraint Adherence and Specification Refinement (RCASR) taxonomy, together with a semi-automatic pipeline based on Legal-S-BERT retrieval, signal-word extraction, and a syntactic decomposition step that uses part-of-speech tagging and dependency parsing to split each constraint into responsibility (subject), task (verb), and data (object) slots. Their best configuration reaches 70.6 % recall but requires manually tuned cosine thresholds and operates on signal-word-filtered sentences, a choice that the same authors note is unsuitable for free-form policy text where normative diction is rare. LLMs (Large Language Models) promise a template-free, fully automated alternative, yet their behaviour on the structured *deviation typing* task that RCASR demands is not well characterised on real-world policies.

This paper presents an end-to-end LLM pipeline that replaces every manually-supervised step of Sai et al. [2023] with a zero-shot or retrieval-augmented LLM call. The four-step flow is as follows. (i) A contextual extractor reads each policy sentence together with a ± 5 -sentence window of surrounding text and decides whether the sentence describes a data-protection obligation. (ii) A separate signal-word filter on the regulatory side yields 279 normative GDPR (General Data Protection Regulation) sentences from Articles 5–43; both sides are embedded with Legal-BERT using a ± 1 -sentence window on the policy side, and for each of the 279 GDPR constraints the top-20 policy passages by cosine similarity are retrieved as candidates. (iii) An LLM-as-judge receives all 20 candidates for a constraint in a single prompt and returns the subset that substantively realises it. (iv) A zero-shot classifier assigns one of six RCASR deviation types to each confirmed pair; GDPR constraints for which the judge returned an empty set are emitted directly as RCASR3 (*constraint_coverage*) deviations. The same prompts run against either OpenAI GPT-4o-mini or a fully local Qwen-3.5-9B served via Ollama, so the system can be deployed on-premise without an external API. We evaluate the pipeline against gold standards of seventeen controlled deviations injected into three real privacy policies (Hetzner, Zalando, Trade Republic) spanning IT infrastructure, fashion e-commerce, and fintech.

The remainder of this paper is structured as follows. Section 2 positions the work against related literature. Section 3 describes the four pipeline steps and the design rationale for each. Section 4 reports per-use-case and per-type results. Section 5 documents the three biggest unresolved challenges and the partial mitigations attempted. Section 6 sketches the E2E compliance workflow and future directions, and Section 7 concludes.

2. Related Work

Research on regulatory compliance checking traces back to Sadiq [2013] and Hashmi et al. [2018], who established that verifying normative text against business processes is an open research problem rather than a solved engineering task. Amaral et al. [2023] then demonstrated that NLP pipelines can reach 84–94 % accuracy when checking GDPR clauses in Data Processing Agreements, validating constraint-pair extraction as a viable approach. Our work extends this direction to free-form company privacy policies, which use considerably less standardised language than the DPAs studied by Amaral et al.

The immediate baseline is the RCASR framework of Sai et al. [2023], which introduces a six-type deviation taxonomy (constraint coverage, severity, execution style, negation, responsibility, data) and a

semi-automatic pipeline pairing GDPR signal-word sentences with policy sentences via Legal-S-BERT cosine retrieval. Their best configuration reaches 70.6 % recall on GDPR Articles 5–91 but relies on (a) manually tuned similarity thresholds, (b) POS-based decomposition into responsibility/task/data components, and (c) signal-word extraction on both sides, a filter the same authors note is too sparse for company policies. We adopt the RCASR taxonomy (types RCASR3–RCASR8) and the constraint-pair architecture verbatim while replacing all three semi-automatic components.

On the embedding side, Chalkidis et al. [2020] pre-trained BERT on 12 GB of legal text and showed consistent improvements on legal NLP tasks over general-domain BERT. We use `nlpaueb/legal-bert-base-uncased` as the retrieval backbone, applying mean-pooling over token embeddings via the `SentenceTransformer` library [Reimers and Gurevych, 2019] to obtain a single vector per sentence. The siamese fine-tuning stage of Sentence-BERT, which calibrates cosine similarity for semantic relevance, is not performed here, so we do not claim retrieval quality comparable to Sai et al.’s Legal-S-BERT; Legal-BERT mean-pooling serves instead as a coarse pre-filter that narrows each GDPR constraint to a top- k candidate set, with the semantic matching work delegated to the LLM-as-judge in Step 3.

Wang et al. [2023] provided early evidence that zero-shot prompts on modern LLMs match or exceed task-specific classifiers for short text spans; we apply this twice in our pipeline, once for the binary obligation-extraction question and once for the six-way deviation typing. For the matching step we follow Zheng et al. [2023], who showed that strong LLM judges reach > 80 % agreement with human annotators on pairwise alignment judgements—the same setting that arises when deciding whether a policy passage substantively realises a GDPR constraint. Our contribution is to combine the RCASR taxonomy with these capabilities, replacing every manually-supervised step with a zero-shot or retrieval-augmented LLM call.

3. Pipeline Architecture and Design Choices

The detection pipeline consists of four sequential steps fed by two parallel input streams, sketched in Figure 1. The GDPR stream reuses the signal-word filter of Sai et al. [2023] unchanged; the policy stream replaces it with a contextual LLM extractor. The two streams meet at a Legal-BERT retrieval step that proposes the top-20 candidate matches per GDPR constraint; an LLM-as-judge then confirms or rejects each candidate; surviving pairs are classified into one of six RCASR deviation types or marked compliant. The remainder of this chapter documents each step and the design rationale behind it.

3.1. Data Inputs

Regulatory side. We use the official EUR-Lex XHTML edition of the GDPR [European Parliament and Council of the European Union, 2016], restricted to Articles 5–43. Articles 1–4 (glossary and scope) and 44–99 (transfers, enforcement, institutions) are excluded because they contain no operational obligations on the controller or processor, matching the operational scope of Sai et al. [2023]. The XHTML is converted to plain text and split into sentences; sentences containing one of the signal words `shall`, `should`, or `must` are retained, yielding 279 GDPR constraints. The signal-word filter is justified on the regulatory side because the GDPR drafts every obligation in consistent normative diction.

Policy side. Three real privacy policies serve as use cases: Hetzner Online GmbH (IT infrastructure, retrieved 2026-04-22), Zalando SE (fashion e-commerce, July 2025 version), and Trade Republic Bank GmbH (retail fintech, version 2025-06-23). The pipeline does not run against the original policies but against modified versions into which controlled deviations were injected; the gold-standard design is described in §3.1.

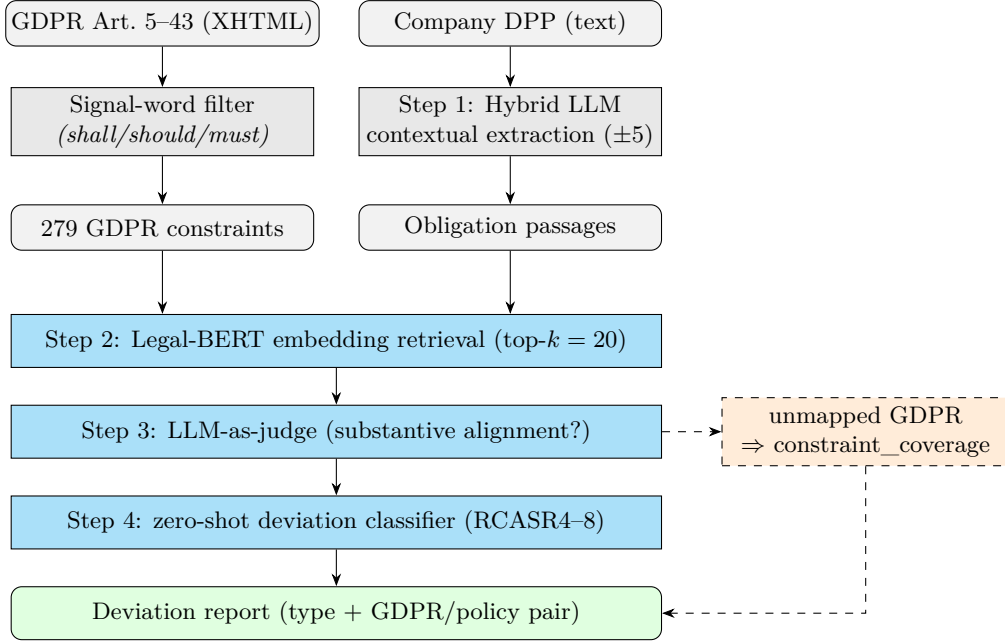


Figure 1 End-to-end detection pipeline. Gray boxes are pre-processing steps; blue boxes are the four LLM-augmented stages (§3.1–§3.1); the dashed branch marks GDPR constraints with no surviving policy match, which are emitted as RCASR3 (*constraint_coverage*) deviations directly without entering Step 4.

Step 1: Hybrid LLM policy extraction. The same signal-word filter applied to the Hetzner policy yields only 28 sentences, a corpus too sparse to plausibly match against 279 GDPR constraints, and a quantitative confirmation of the warning by Sai et al. [2023] that company policies do not adopt the normative diction of regulatory texts. Step 1 therefore replaces the filter with a contextual LLM extractor. For each candidate sentence in the policy, the extractor is given a sliding ± 5 -sentence window of surrounding context and asked, via a structured prompt, whether the sentence *describes a data-protection obligation, right, or processing practice*. The same prompt is used regardless of the underlying model; all results reported in this paper use OpenAI GPT-4o-mini [OpenAI, 2024], and Qwen-3.5-9B served locally via Ollama [Team, 2025] is supported as a drop-in fully-on-premise replacement. This design is supported by Wang et al. [2023], who show that zero-shot prompts on instruction-tuned LLMs can serve as strong classifiers on short text spans without task-specific training. Discarding the signal-word baseline was an explicit choice rather than an oversight: it was retained as a calibration check on the regulatory side where it provably works, and replaced only where it provably does not.

Step 2: Legal-BERT embedding retrieval. Both GDPR constraints and surviving policy passages are embedded into the same vector space with `nlpaueb/legal-bert-base-uncased` [Chalkidis et al., 2020], exposed through the `SentenceTransformer` library with mean-pooling over the final hidden states [Reimers and Gurevych, 2019]. Using a domain-adapted encoder rather than general-domain BERT matters here because legal equivalence often hides behind paraphrase: “the controller shall inform the data subject” and “we will notify you” share little surface vocabulary but high legal equivalence, and Legal-BERT places them substantially closer than the general-domain model does. For each GDPR constraint, the top- $k = 20$ policy passages by cosine similarity are forwarded to the judge in Step 3. The context window used for embedding the policy side is ± 1 sentence rather than the ± 5 window used for Step 1: Legal-BERT’s 512-token limit truncates ± 5 windows in practice, and a fixed truncation point near the centre sentence yielded measurably worse retrieval than a deliberately narrow window. The value $k = 20$ was selected after preliminary runs with $k \in \{5, 10\}$: both gave insufficient recall on the gold-standard deviations in the candidate set, and the eventual judge step (§3.1) makes a larger k cheap because the judge prompt batches all 20 candidates into a single LLM call.

Step 3: LLM-as-judge. Cosine similarity over Legal-BERT alone is not sufficient to declare a policy passage a substantive realisation of a GDPR constraint: the top-1 cosine match is often topically related but legally inert, and the threshold that separates “related” from “substantive” varies by GDPR article. Sai et al. [2023] addressed this with a manually tuned global threshold. We instead delegate the decision to an LLM-as-judge in the sense of Zheng et al. [2023], who report $> 80\%$ agreement with human annotators on pairwise judgements of this shape. For each GDPR constraint, the judge receives the constraint together with all 20 candidate passages in a single prompt and returns a JSON list of indices whose passage substantively realises the constraint. This design keeps the call budget tractable: exactly 279 LLM calls are issued in total, compared to roughly $279 \times 60 \approx 16,000$ naïve pairwise calls if every candidate were judged separately. Constraints for which the judge returns an empty list are routed around Step 4 directly to a special outcome, RCASR3 (*constraint_coverage*), because they correspond to obligations the policy never substantively addresses.

Step 4: Deviation classification. Each pair confirmed in Step 3 is handed to a zero-shot classification prompt that asks the model to assign one of six labels: RCASR3 (*constraint_coverage*) is the side-channel from Step 3; RCASR4–RCASR8 (*severity, execution_style, negation, responsibility, data*) are decided here. The prompt names each type, gives a one-sentence definition, and asks the model to either select exactly one type or to return **compliant**. Conceptually, Sai et al. [2023] factorise a constraint into responsibility/task/data parts via POS and dependency parsing, then check for occurrence mismatch per slot; this works on highly templated text but becomes brittle on free-form privacy policies where the responsible actor (e.g. “our support team”) is implicit and the task verb varies. Wang et al. [2023] found zero-shot LLMs robust to exactly this kind of paraphrastic variation. Empirically, the same model that performs the extraction and judging steps performs the classification without task-specific fine-tuning. To guard against model self-confirmation bias, the gold-standard deviations injected into the policies are generated by a different model family (Claude Sonnet 4.6) than the model performing detection (GPT-4o-mini in the reported runs, Qwen-3.5-9B in the local-only deployment). The detection and generation models therefore share neither weights nor prompt artefacts in either configuration.

Gold standard design. A reliable evaluation requires deviations whose ground truth is known. The original Hetzner, Zalando, and Trade Republic policies are used as base text; into each policy we manually inject seventeen deviations of known RCASR type. Three deviations are introduced for every RCASR4–RCASR8 type (*severity, execution_style, negation, responsibility, data*); two more are introduced for RCASR3 (*constraint_coverage*) by removing previously covered clauses. The original plan called for eighteen deviations per use case (three per type $\times$ six types), but one RCASR3 entry per policy targeted GDPR Art. 77 (right to lodge a complaint), which falls outside the Art. 5–43 scope used by the pipeline. Because no GDPR constraint exists for those entries in the regulatory index, they would have been structurally undetectable and were therefore removed. The distribution is summarised in Table 1.

Each injected deviation is logged in a JSON manifest (*gold_standard/*_deviation_manifest.json*) with its type, the modified policy span, and a brief description, so the evaluator can map every prediction back to the ground-truth entry without ambiguity.

4. Evaluation

4.1. Experimental Setup

We evaluate the pipeline against the gold standards introduced in §3.1 on three use cases chosen for sector diversity: Hetzner Online GmbH (IT infrastructure), Zalando SE (fashion e-commerce), and Trade Republic Bank GmbH (retail fintech). Each use case is evaluated independently: the pipeline is run against the modified policy, predictions are matched to entries in the corresponding deviation manifest, and precision, recall, and F_1 are computed per use case and per RCASR type. The numbers reported below were produced with OpenAI GPT-4o-mini [OpenAI, 2024] as the LLM

Table 1 Gold standard: number of injected deviations per type and use case. Three deviations per type are inserted for RCASR4–RCASR8; two for RCASR3 (*constraint_coverage*) following removal of one Art. 77 entry per policy that was outside the pipeline’s Art. 5–43 scope.

Deviation type	Hetzner	Zalando	Trade Republic
RCASR3: <i>constraint_coverage</i>	2	2	2
RCASR4: <i>severity</i>	3	3	3
RCASR5: <i>execution_style</i>	3	3	3
RCASR6: <i>negation</i>	3	3	3
RCASR7: <i>responsibility</i>	3	3	3
RCASR8: <i>data</i>	3	3	3
Total	17	17	17

backend; the same pipeline runs unchanged against Qwen-3.5-9B served locally via Ollama [Team, 2025], which is the configuration used when an external API is not permitted (for example during the on-premise compliance review for which the tool is ultimately intended). Embeddings are computed with `nlpaueb/legal-bert-base-uncased`. The entire pipeline is containerised and reproducible from a fresh checkout via `docker compose up`.

One methodological remark on the metrics is in order. Recall is the primary reliability indicator in this setup because the gold standard covers only the seventeen deliberately injected deviations; any other predicted deviation is counted as a false positive even though some of those predictions may identify genuine but undocumented misalignments in the original policy text. The reported precision is therefore a lower bound on the true precision of the system.

4.2. Results

Table 2 reports overall precision, recall, and F_1 per use case; Table 3 breaks the aggregate down by RCASR type.

Table 2 Overall precision, recall, and F_1 per use case. Aggregate figures are micro-averaged over the 51 ground-truth deviations (17 per use case $\times$ 3 use cases).

Use case	Precision	Recall	F_1
Hetzner Online GmbH	31.0 %	52.9 %	39.1 %
Zalando SE	25.0 %	47.1 %	32.7 %
Trade Republic Bank GmbH	25.0 %	47.1 %	32.7 %
Aggregate	26.9 %	49.0 %	34.7 %

Table 3 Aggregate precision, recall, and F_1 by deviation type, micro-averaged over all three use cases.

Type	Precision	Recall	F_1
RCASR3: <i>constraint_coverage</i>	50.0 %	16.7 %	25.0 %
RCASR4: <i>severity</i>	30.8 %	44.4 %	36.4 %
RCASR5: <i>execution_style</i>	50.0 %	66.7 %	57.1 %
RCASR6: <i>negation</i>	20.0 %	77.8 %	31.8 %
RCASR7: <i>responsibility</i>	22.2 %	22.2 %	22.2 %
RCASR8: <i>data</i>	22.7 %	55.6 %	32.3 %

4.3. Discussion

Hetzner’s headroom is structural, not methodological. Hetzner scores higher than the other two use cases on every metric, but the difference is not driven by the detection pipeline. The Hetzner policy contains roughly 56 obligation sentences against 80–100 for Zalando and Trade Republic, and its deviations concentrate around Articles 12–22 (data subject rights), where the GDPR text is most explicit. A shorter, more focused policy presents fewer candidate passages per GDPR constraint, which directly reduces the retrieval noise that Step 3 must filter out.

execution_style is the best-detected type. Type RCASR5 reaches $F_1 = 57.1\%$, well ahead of the other types. These are deviations in how a constraint is realised (“*we will respond within 30 days*” versus “*response time may vary*”), and the contrast tends to be lexically and semantically distinct enough that both the retrieval and the classifier latch onto it cleanly.

responsibility is the hardest. Type RCASR7 lands at $F_1 = 22.2\%$, the lowest of the six. The classifier struggles when the policy uses implicit actors (“our customer support team handles these requests”) against the GDPR’s explicit named roles (“the controller”, “the DPO (Data Protection Officer)”). The judge in Step 3 accepts the pair as substantively aligned, correctly, because the policy does cover the obligation, but the classifier in Step 4 fails to distinguish a genuine responsibility shift from a paraphrase.

Precision is a lower bound. The classifier flags roughly 40% of confirmed pairs as deviations. For a policy that has only seventeen known deviations among hundreds of candidate pairs, the realistic base rate is closer to 3–5%, and the zero-shot prompt has no mechanism to learn this base rate. The recorded precision is therefore a lower bound: an unknown but plausibly non-trivial share of flagged deviations may be genuine, undocumented misalignments in the original policy. Investigating these without a secondary gold standard is outside this paper’s scope.

Comparison to Sai et al. [2023]. Their best configuration on GDPR reaches 70.6% recall using Legal-S-BERT with contrastive fine-tuning and manually tuned cosine thresholds. Our fully automated pipeline reaches 49.0% recall on a different gold standard. A direct head-to-head comparison is not possible, since the two gold standards differ in size, distribution, and construction method, but the recall gap is informative: it quantifies roughly the cost of removing all manual tuning steps from the pipeline.

5. Challenges and Tried Solutions

Granularity mismatch between GDPR and policy. The 279 GDPR constraints decompose obligations at sentence granularity, whereas company policies typically address the same topic with a single paragraph or even a single sentence enumerating multiple articles (“rights under Articles 15, 16, 20 may be exercised”). The judge in Step 3 then accepts the enumeration as substantively realising each of the cited articles, which is partially correct but inflates the candidate pair count and damages precision once Step 4 begins flagging deviations. We tried two mitigations: (i) tightening the judge prompt to require “a specific, actionable obligation” rather than mere topical overlap, and (ii) adding a cosine pre-filter that drops candidates below 0.45 similarity. Both produced marginal gains because the root cause is structural: enumeration sentences *are* legally substantive realisations, just not constraint-level ones.

Classifier without base-rate prior. The Step 4 classifier flags roughly 40% of confirmed pairs as deviations, against an estimated real-world base rate of 3–5%. The zero-shot prompt has no mechanism to encode the prior that most pairs in a published company policy are compliant. We experimented with prompt-level calibration by adding the statement “*the majority of pairs you will*

see do not contain deviations; only report a deviation when the mismatch is concrete and verifiable” to the system prompt. Precision rose modestly but recall fell by a similar margin, leaving F_1 unchanged. Closed-form calibration, either via few-shot examples with the correct class distribution or via a separate calibration head on top of the classifier, remains an open problem.

6. E2E Integration Architecture and Future Work

The detection pipeline as presented covers the automated core of a compliance workflow but is not yet a tool a compliance officer can pick up unaided. Figure 2 situates the implemented pipeline (blue, centre) within the broader workflow it would slot into in production. The dashed box marks the single missing component: an ingestion and review front-end that lets a non-technical user upload a new policy and review the resulting deviation report.

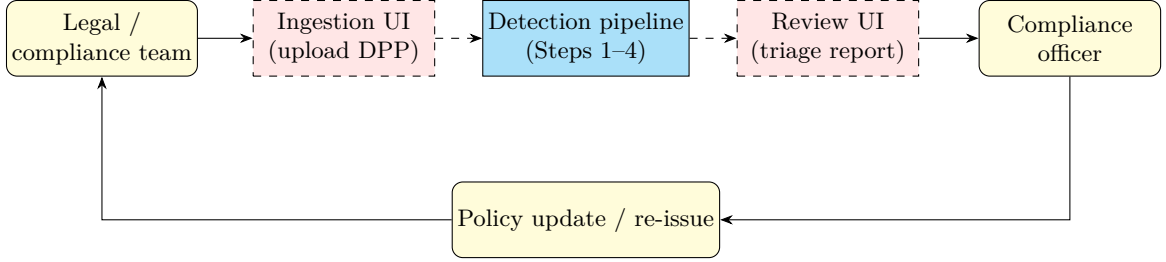


Figure 2 Conceptual end-to-end compliance workflow. The blue box is the detection pipeline implemented in this paper, reproducible via `docker compose up`. The dashed red boxes mark the ingestion and review interfaces that would close the loop from an uploaded policy document to a triaged deviation report and a re-issued policy.

Future work. Three directions follow naturally from the open challenges in Section 5. First, *base-rate calibration* for the Step 4 classifier, either via few-shot examples drawn from a separate holdout policy or via an explicit calibration head, is the cheapest intervention with the largest expected precision gain. Second, *article-specific judge prompting* together with a contrastive Legal-S-BERT fine-tuned on legal alignment pairs should reduce the enumeration-sentence false matches that drive the granularity-mismatch problem. Third, few-shot exemplars per deviation type are the most direct lever for the two worst-performing types, *responsibility* ($F_1 = 22.2\%$) and *data* ($F_1 = 32.3\%$), because both suffer primarily from category confusion rather than from retrieval failure.

7. Conclusion

We have presented a fully automated pipeline for detecting RCASR-typed deviations between GDPR Articles 5–43 and company data protection policies, replacing the semi-automatic NLP components of Sai et al. [2023] with a contextual LLM extractor, a Legal-BERT retrieval step, an LLM-as-judge for substantive matching, and a zero-shot deviation classifier. Evaluated against gold standards of seventeen controlled deviations injected into the real privacy policies of Hetzner, Zalando, and Trade Republic, the pipeline reaches an aggregate F_1 of 34.7% at recall 49.0%, with *execution_style* the best-detected type ($F_1 = 57.1\%$). To the best of our knowledge this is the first fully automated, off-the-shelf LLM pipeline evaluated against RCASR-typed gold standards on multiple real-world company policies. The two clearest paths to improving practical precision are base-rate calibration in the classifier prompt and few-shot exemplars for the responsibility and data deviation types; both are inexpensive and structure-preserving and are left to future work.

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